MOSU2: Long-range Navigation System with Multi-modal Scene Understanding

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Abstract

Reliable long-range outdoor navigation requires mobile robots to assess terrain traversability, interpret social and traffic context, and respond to the evolving behavior of surrounding road users. These requirements are closely coupled at crossings and intersections, where the appropriateness of a trajectory depends on both contextual cues and road-user motion. Building on MOSU, we present MOSU2, an integrated navigation framework that combines route guidance, multi-modal traversability analysis, and motion-grounded contextual reasoning. The framework maintains a local map of tracked road users, providing geometric and dynamic information to support the interpretation of traffic interactions. A specialized multi-agent architecture organizes this reasoning into two stages: a context agent interprets RGB observations to determine whether further investigation is needed, and only when it is, a decision agent retrieves and analyzes road-user states to select an appropriate trajectory. By grounding contextual interpretations in measured motion, the framework supports decisions involving approaching, stopping, and potentially yielding vehicles, while incorporating traffic signals and pedestrian activity. Evaluation focuses on signalized and unsignalized crossings involving pedestrians, cyclists, and vehicles, examining navigation reliability, unnecessary delay, and the contribution of each reasoning component. [Insert the principal quantitative findings and comparisons with the original MOSU, conventional dynamic planning, and a single-agent reasoning baseline.]

Keywords

Long-range outdoor navigation; social navigation; traversability analysis.

Introduction

Long-range robot navigation has advanced significantly in recent years Gao et al. (2019); Liang (2025). This task requires mobile robots to perceive and reason about heterogeneous environments to reach distant destinations safely, efficiently, and in a socially compliant manner Gul et al. (2019); DeSouza and Kak (2002); Mavrogiannis et al. (2023). It plays a crucial role in applications such as logistics Alverhed et al. (2024); Chen et al. (2021), service robotics Zemke et al. (2020); Berezina et al. (2019), and search and rescue Davids (2002); Kalita and Thangavelautham (2020).

Despite these advances, long-range navigation remains challenging in diverse and unstructured environments, where robots with different mobility capabilities Liang et al. (2025); Eder et al. (2023); Wellhausen and Hutter (2021) encounter varying terrain Weerakoon et al. (2023); Sathyamoorthy et al. (2022), irregular obstacle motion Wijayathunga et al. (2023), and complex traffic and social interactions Liang et al. (2026); Payandeh et al. (2025); Kong et al. (2025). For example, a robot must identify terrain that is compatible with its physical capabilities and, before entering the crosswalk, determine whether an approaching vehicle is yielding. These situations require the robot to coordinate global route guidance, local traversability assessment, and contextual reasoning while accounting for the evolving behavior of surrounding road users.

Long-range navigation requires both global guidance to a distant goal and accurate, real-time local scene understanding so that the robot can move safely and efficiently, as illustrated in Fig. 1. Existing navigation systems use global SLAM-based maps to support localization and path planning Gasparetto et al. (2015); Ozturk et al. (2022), while approaches such as ViNT combine learned visual navigation with topological graph-based planning Shah et al. (2023). These representations provide useful global structure, but constructing and maintaining detailed maps over large environments can be costly, and environmental changes can invalidate previously traversable routes. Moreover, global guidance does not necessarily capture the terrain conditions or temporary obstructions encountered during execution. The challenge is therefore to maintain progress toward a distant goal while reconciling global guidance with limited local observations. Our framework uses publicly available route information and GPS localization for coarse guidance, while

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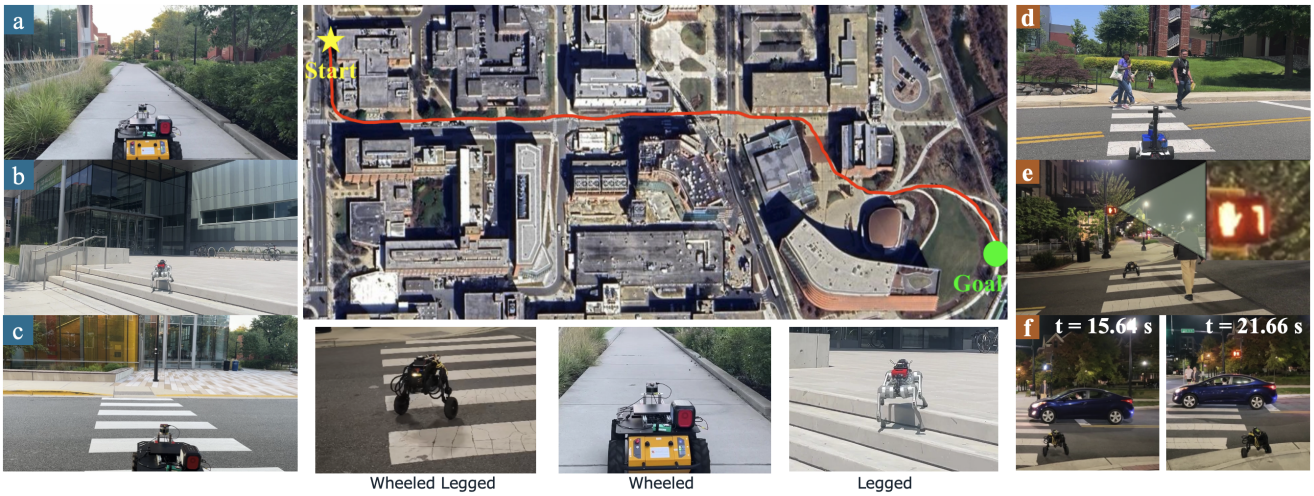


Figure 1. MOSU2 supports long-range outdoor navigation across wheeled-legged, wheeled, and legged robot platforms (bottom row). The satellite view illustrates a route from the start (yellow star) to the goal (green circle). The left-side examples highlight traversability challenges: (a) distinguishing sidewalks from off-road terrain, (b) accounting for differences in robot mobility, and (c) identifying crosswalks for street crossing. The right-side examples highlight social and traffic challenges: (d) interacting with moving agents, such as pedestrians, (e) interpreting traffic signals, including pedestrian signals with countdown displays, and (f) resolving conflicting traffic cues when vehicles violate traffic rules and obstruct the robot's path. MOSU2 combines global route guidance and robot-specific traversability assessment with two-stage reasoning that retrieves local geometric and dynamic information to interpret these conditions and select appropriate motion.

onboard perception determines locally traversable motion without requiring a detailed prebuilt metric map.

Another challenge is to assess traversability according to the robot's morphology and motion capabilities, as illustrated by the terrain variations in Fig. 1 (a-c). Geometric methods maintain local elevation maps to represent terrain shape and obstacles Fankhauser et al. (2018), while semantic methods estimate traversability to distinguish areas that have similar geometry but different implications for navigation Kim et al. (2024). Learning-based approaches generate local actions or candidate trajectories directly from onboard observations Liang et al. (2024a); Shah et al. (2023); Sridhar et al. (2024). Nevertheless, local-map reconstruction remains limited by sensor coverage and occlusion, particularly around corners or in crowded environments, while learned semantic predictions are constrained by the training distributions Kim et al. (2024). Our framework combines learned trajectory generation with geometric and robot-specific semantic traversability constraints to evaluate candidate motions.

Beyond physical traversability, robots operating in shared public spaces must understand social cues and traffic conditions, as illustrated in Fig. 1 (d-e). Human gestures, pedestrian signals, countdown displays, and vehicle stopping behavior convey information whose meaning depends on the surrounding situation and its evolution over time. Recent approaches use vision-language models (VLMs) to incorporate contextual knowledge into social navigation Song et al. (2025a); Payandeh et al. (2025). However, methods interpreting individual images can miss the temporal evidence needed to analyze ongoing motion from changes in behavior, and qualitative descriptions do not necessarily provide reliable metric information for planning Kong et al. (2025). Our approach grounds visual-context reasoning in geometric and dynamic information retrieved from a local map.

Another challenge is translating these complementary, and sometimes conflicting, observations into an executable navigation decision, as illustrated in Fig. 1 (f). For instance, if the pedestrian signal indicates that crossing is permitted but a vehicle is slowly approaching without yielding, the robot must be able to prioritize these conflicting cues to make a safe navigational decision. Existing approaches integrate contextual information through behavioral cost maps Weerakoon et al. (2025) or VLM-based selection among paths generated using geometric and human-motion information Fang et al. (2026). Nevertheless, ambiguity among concurrent activities and missing temporal context can still lead to inappropriate prioritization of social cues Fang et al. (2026). The remaining challenge is to identify which observations matter for the current decision and reconcile their implications with physical feasibility and progress toward the goal. Our approach organizes contextual interpretation, geometric and dynamic analysis, and trajectory selection into specialized agents that exchange information relevant to the navigation decision.

Our previous work, MOSU Liang et al. (2026), established a long-range navigation framework that combines coarse GPS-based routing with multimodal local scene understanding. It generates candidate trajectories from LiDAR and robot-motion observations, applies semantic traversability constraints using RGB images, and uses a VLM to assess the social appropriateness of candidate trajectories for various robot types. However, although its trajectory generator incorporates temporal sensor inputs, its VLM-based social reasoning does not explicitly associate visual cues with metric geometry and motion history. Consequently, the system has limited understanding of critical interactions that emerge over time, such as whether a vehicle or a bicycle is approaching, stopping, or yielding. In addition, its predefined prompting and trajectory-scoring pipeline lacks an explicit mechanism for selecting and retrieving additional

scene information when the current semantic and geometric information is insufficient to interpret an interaction. These limitations motivate a tighter connection between contextual reasoning, measured road-user states, and navigation decisions.

To address these limitations, we present MOSU2, an extension of MOSU that combines information retrieval and two-stage reasoning for long-range navigation. MOSU2 allows VLMs to retrieve geometric and dynamic information from a local occupancy grid map and interpret it together with social and traffic cues from RGB observations. We also propose a two-stage multi-agent architecture that coordinates this process by 1. identifying relevant information for navigation, and 2. retrieving and interpreting supporting evidence, and selecting trajectories within MOSU's routing and traversability framework. In most of the outdoor real-world scenarios, we don't need sophisticated understanding of social and traffic conditions, and thus can be quickly processed by the first stage; but if there is anything requires investigating more investigation, the second-stage scene understanding could provide more details and make the navigation decision. Our contributions beyond MOSU are:

- We introduce a method that connects VLM-based contextual understanding with geometric and dynamic information represented in a local map. The method enables VLMs to retrieve relevant map information and relate surrounding geometry and motion to observed social and traffic cues.
- We develop a two-stage architecture that coordinates information gathering and navigation decisions. Specialized agents 1) identify if there is more information needed for the current situation, 2) analyze visual context, geometric and dynamic information of the scene, and combine their findings with traversability assessments and route guidance to select an appropriate trajectory.

Related Work

Long-range Outdoor Navigation and Traversability

Long-range outdoor navigation requires robots to reconcile guidance toward a distant destination with locally feasible motion. Several approaches address this problem without requiring a detailed global metric map. ViNT Shah et al. (2023) learns a general-purpose visual navigation policy from heterogeneous robot datasets and combines it with topological planning for extended navigation. NoMaD Sridhar et al. (2024) further unifies goal-directed navigation and exploration through a goal-conditioned diffusion policy. These approaches demonstrate that coarse global information and learned local navigation can support long-range autonomy without reconstructing the entire environment in metric detail.

Recent methods improve outdoor navigation through broader training data and longer planning horizons. CityWalker Liu et al. (2025) learns urban navigation policies from large-scale walking and driving videos, using observation and trajectory histories to predict future motion. Long Range Navigator Schmittle et al. (2025) predicts promising frontier directions from egocentric images, extending the effective horizon of local navigation systems beyond their metric maps. These methods address complementary challenges in generalization and goal-directed progress. However, their navigation interfaces do not explicitly request additional metric evidence to resolve a particular social or traffic ambiguity. This distinction motivates our context agent, which uses contextual reasoning to identify when route-following requires further investigation of the surrounding scene.

Local traversability provides another essential component of outdoor navigation. ViPlanner Roth et al. (2024) combines depth and semantic observations to generate paths that account for terrain-dependent traversability, while MTG Liang et al. (2024a) generates diverse candidate trajectories with objectives that encourage traversability and coverage. However, physical traversability alone does not determine whether a motion is appropriate. For example, a traversable crossing may require waiting for traffic, and an accessible pathway may be temporarily restricted. Our previous MOSU framework Liang et al. (2026) integrates candidate generation, geometric and semantic assessment, and visual-context evaluation with GPS-based route guidance. MOSU2 retains this navigation foundation while introducing additional reasoning over social, traffic, and temporary environmental conditions.

Social and Traffic-aware Navigation

Social navigation extends collision avoidance by considering how robot motion affects nearby people and whether it conforms to contextual expectations. ORCA Van Den Berg et al. (2011) computes reciprocal collision-avoidance actions using the relative motion of neighboring agents. Learning-based approaches, such as the attention-based crowd-navigation policy of CrowdSteer. Liang et al. (2021), additionally model human-robot and human-human interactions to anticipate crowd behavior. These methods explicitly account for motion, but their state-based formulations do not directly represent the meaning of gestures, traffic signals, or temporary restrictions.

Vision-language models provide a complementary means of interpreting such contextual information. VLM-Social-Nav Song et al. (2025a) uses visual-language reasoning to score socially appropriate directions and speeds for a downstream navigation system. Social-LLaVA Payandeh et al. (2025) develops navigation-oriented visual reasoning through supervised social-scene understanding, while AutoSpatial Kong et al. (2025) improves spatial reasoning through structured grounding and hierarchical visual question answering. These approaches broaden the range of social situations that navigation systems can handle. However, understanding an interaction from an individual image remains limited when its meaning depends on dynamic information. AutoSpatial explicitly identifies temporal understanding as an open challenge, including cases in which human intentions become apparent only across successive observations. This limitation motivates combining image sequences with measured spatial and motion information.

Agentic Robotics and Adaptive Reasoning

Robotic planning agents increasingly use external representations and tools to connect language reasoning with observations of the physical environment. NLMap Chen et al. (2023) constructs an open-vocabulary scene representation that allows a language-model planner to query object availability and location before generating a plan. SayPlan Rana et al. (2023) uses semantic search over hierarchical 3D scene graphs, together with conventional path planning and feedback-based replanning, to support large-scale mobile-manipulation tasks. More recently, the SuperNav preprint Zhang et al. (2026) organizes navigation around a pretrained multimodal model with tool access, task-progress tracking, and context management. These approaches motivate separating information acquisition from planning while maintaining an explicit connection between the two.

The computational cost of foundation-model reasoning has also motivated architectures that separate deliberation from rapid execution. DualVLN Wei et al. (2026) combines a VLM that predicts intermediate goals with a lightweight diffusion policy for local motion. Hydra-Nav Wang et al. (2026) adaptively switches between deliberative reasoning and reactive execution for object-goal navigation. Particularly relevant to outdoor deployment, the Walk With Me preprint **?** combines public-map routing, high-level VLM scene assessment, and low-level VLA trajectory generation. Its high-level model assesses whether the robot should proceed or stop and wait in situations such as crossings and crowded scenes. These methods demonstrate the relevance of hierarchical and adaptive reasoning, so the use of two stages alone does not distinguish MOSU2.

Our Approach

Our approach consists of two parts: the navigation framework inherited from MOSU Liang et al. (2026) and the extensions introduced in MOSU2. MOSU decomposes long-range navigation into routing and local trajectory generation and evaluation. The routing module uses QGIS and GPS data to provide global guidance for long-range navigation, while trajectory generation and evaluation combine multi-modal sensor observations to generate candidate trajectories, assess traversability costs, and evaluate compliance with social and traffic rules from RGB perception. However, because MOSU does not use the geometric and dynamic information of the social and traffic conditions, it behaves conservatively by either stopping the robot whenever pedestrians or bicycles are nearby, regardless of how they move relative the robot. Building on this framework, MOSU2 introduces a VLM-retrievable local cost map that provides geometric and dynamic information relevant to the observed scene context. Real-world environments are complicated and contains potential hazardous conditions that cannot be easily detected by traversability analysis, such as dangerous construction sites or road closures marked only by subtle signs. To handle these challenges, we design a two-stage multi-agent reasoning architecture. In the first stage, the agent handles navigation decisions using the available observations and trajectory assessments with MOSU's route guidance and trajectory assessments and determines whether further investigation is needed. When additional information is required, the decision agent in second stage retrieves relevant map information, coordinates visual-context interpretation with geometric and dynamic analysis, and combines the resulting evidence to select an appropriate trajectory.

Global Direction From Routing

We retain the routing strategy of MOSU Liang et al. (2026) to provide coarse global guidance for long-range navigation. Given the robot's current GPS position and a target location, the routing module obtains a high-level route from a public routing service, such as Google Maps or OpenRouteService, and represents it as a sequence of intermediate GPS subgoals. These subgoals indicate the intended progression toward the destination, while local trajectory generation and contextual reasoning determine how the robot should move through the observed environment.

Because GPS localization is subject to meter-scale errors Van Diggelen and Enge (2015), subgoal coordinates may not coincide with locally traversable terrain. We therefore use them as directional guidance rather than precise positions to be reached. Following the MOSU implementation, as shown in the Routing block of Fig. 3, consecutive subgoals are spaced approximately $35\,\mathrm{m}$ apart, a distance chosen in relation to the perceptual range of the onboard 3D LiDAR and RGB sensors. This spacing limits reliance on closely spaced GPS coordinates, while local geometric and semantic context assessments guide the robot along traversable terrain. We then transform the GPS subgoal into the robot's odometer frame using the Universal Transverse Mercator projection, and an extended Kalman filter updates this transformation. We have $g = \mathbf{T}\hat{g}$, where $\hat{g}$ contains GPS latitude and longitude, g is the subgoal in the odometer coordinate, and $\mathbf{T}$ is the calculated transformation matrix.

During navigation, the active subgoal is transformed into the robot's coordinate frame using the estimated robot pose. The system monitors the approximate distance to this subgoal and advances to the next one when the distance falls below a predefined threshold of $15\,\mathrm{m}$. Route guidance specifies the desired direction of progress, while local traversability, geometric and dynamic information, and social and traffic context determine the appropriate motion toward it.

Trajectory Generation and Evaluation

Given the noisy subgoal from the GPS due to inherent sensor noise and the complex and unstructured outdoor environment, it is very challenging to generate an optimal trajectory to solve the navigation problem at once. Relying solely on geometric or color information is insufficient to fully understand the environment's traversability (i.e., which areas are safe for the robot to traverse), as shown in Fig. 7. Therefore, our approach integrates both geometric and color information for trajectory generation and evaluation. Classical approaches perform traversability analysis of the environment to provide the constraints for trajectory optimization, but these approaches often require complete perception and are computationally slow and heavy. Learning-based approaches use end-to-end approaches directly convert the high-dimensional perceptive

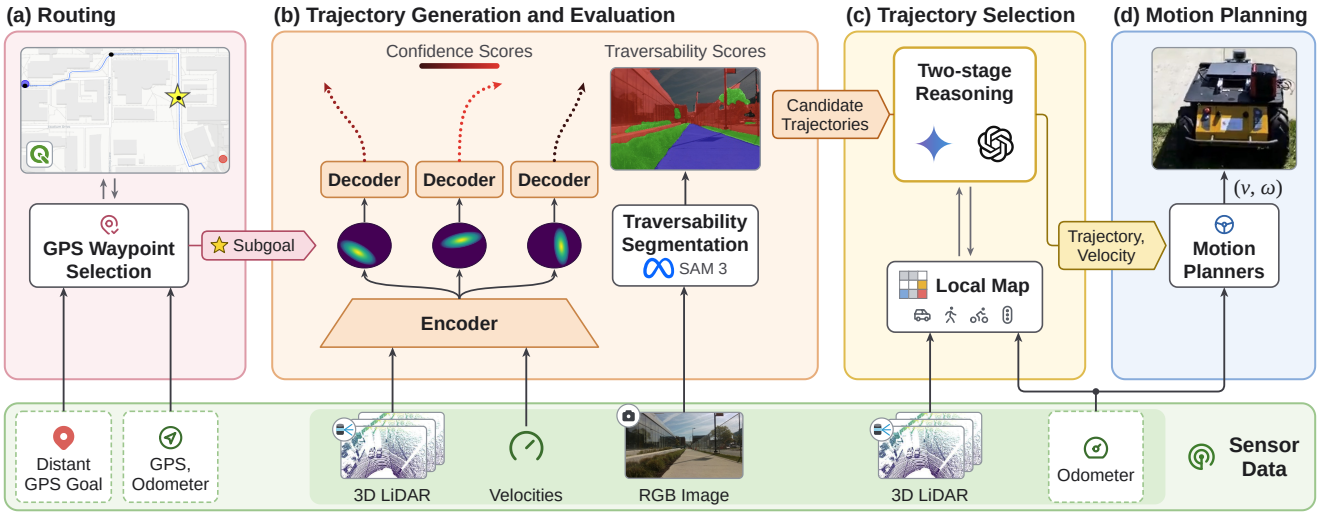


Figure 2. Overview of MOSU2 architecture. Building on MOSU's routing and traversability framework, GPS-based routing provides subgoals, while LiDAR and robot-velocity observations support candidate trajectory generation. RGB-based semantic segmentation provides complementary traversability assessments. MOSU2 introduces a VLM-retrievable local map and two-stage trajectory selection: the first stage screens the current scene and determines if more analysis is required, and the second stage, when invoked, combines recent images with retrieved geometric and dynamic information to select a trajectory and velocity mode. The motion planner converts the selected motion into linear and angular velocity commands (v, ω).

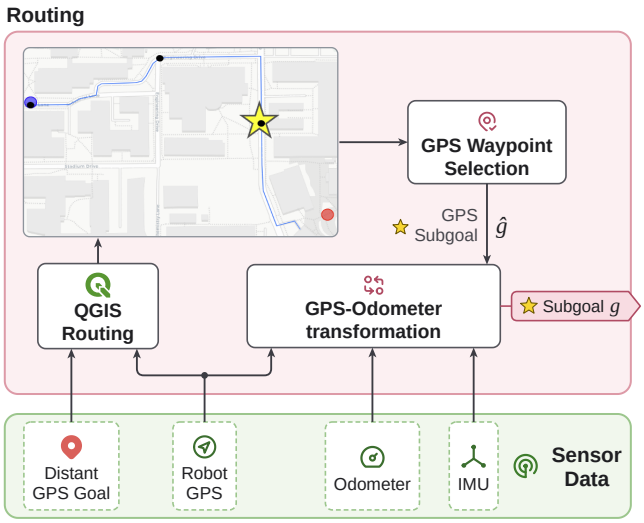


Figure 3. Routing and subgoal transformation. QGIS-based routing uses the robot's GPS location and the destination to obtain a global route, from which an intermediate GPS waypoint $\hat{g}$ is selected. The waypoint is converted to the odometry frame with metric coordinates by aligning the GPS and robot's motion, and outputs the subgoal g. This subgoal provides directional guidance for trajectory selection.

data to low-dimensional trajectory. It's faster, but requires lots of data to cover complex outdoor conditions. Our approach separates this problem into two: we first generate the trajectories to cover all traversable areas in front of the robot, then we can use various models to provide constraints, evaluate the trajectories, and select the best trajectory to move. In this way, we don't need to build an integrated map for trajectory optimization, and we can also leverage the big VLM models' context understanding capability to handle the traffic and social conditions. In this work, we inherit two components from MOSU: (1) CVAE-based multiple trajectory generation, which models the distribution of feasible trajectories based on LiDAR and velocity inputs, (2) semantic segmentation for traversability constraints, which leverages RGB images to identify navigable surfaces.

CVAE-based Multiple Trajectories Generation: This trajectory generation model understands the geometric information of the environment for traversability analysis through 3D Lidar point clouds. However, directly modeling the environment using 3D point clouds is memory-intensive, and segmenting large point cloud is conputationally expensive Guo et al. (2020); Sohail et al. (2024). To address the issue, we adopt MTG Liang et al. (2024a), a lightweight learning-based approach that generates multiple candidate trajectories to cover traversable regions in front of the robot. MTG employs a Conditional Variational Autoencoder (CVAE) Sohn et al. (2015) to model the distribution of feasible trajectories. Given sensor input $\mathbf{x} = \{\mathbf{l}, \mathbf{v}\}$, where $\mathbf{l} \in \mathcal{L}$ represents a sequence of LiDAR observations and $\mathbf{v} \in \mathcal{V}$ denotes the robot's historical velocities, the model generates a set of N candidate trajectories $\mathcal{T} = \{\tau_1, ..., \tau_N\}$. For each trajectory $\tau_n \in \mathcal{T}$, we have the following formulation:

$$p_\theta(\tau_n|\mathbf{x}) = p_\theta(\tau_n|\mathbf{z}_n, \mathbf{c}), \quad \mathbf{z}_n \sim \mathcal{N}(\mu_n, \nu_n), \tag{1}$$

$$\mathbf{c} = f_\theta(\mathbf{x}), \quad \mu, \nu = g_\theta(\mathbf{x}), \tag{2}$$

$$\mathcal{N}(\mu_n, \nu_n) = h_\theta(\mathcal{N}(\mu, \nu), \mathbf{c}) \tag{3}$$

We slightly abuse the notation of θ to represent the parameters of all neural networks. $\mathbf{c}$ is the conditional vector obtained from an encoder network $f_\theta(\cdot)$. $g_\theta(\cdot)$ is the latent encoder network that takes the sensor inputs $\mathbf{x}$ and predicts the parameters, mean μ and variance ν, of a Gaussian distribution $\mathcal{N}(\mu, \nu)$. In our approach, the LiDAR data is processed by PointCNN Li et al. (2018), and the velocities are processed by a sequence of linear layers, as shown in Liang et al. (2024a). Then we linearly transform the distribution to N Gaussian distributions $\mathcal{N}(\mu_n, \nu_n)$ by the learnable linear transformation (neural network) $h_\theta(\cdot)$.

We then sample the latent vector $\mathbf{z}_n$ from the distribution $\mathcal{N}(\mu_n, \nu_n)$ and use it as the input to the trajectory decoder $p_\theta(\cdot)$ to generate trajectory τ_n. The predicted variance ν represents the uncertainty in the latent space and is used to compute a confidence score c_τ for each trajectory. This formulation takes the LiDAR geometric information as input and generates trajectories to cover geometrically traversable areas in front of the robot.

Open-vocabulary Semantic Segmentation for Traversability Analysis: However, the trajectory generation model struggles to accurately detect the boundaries between adjacent traversable and non-traversable areas when they have similar geometric structures (e.g., off-road terrain vs. sidewalks), as shown in Fig.7 (a). To address this limitation, we incorporate semantic information by using SAM3 Carion et al. (2026), an RGB-based semantic segmentation model with open-vocabulary queries. The model segments the image into multiple regions with different queried semantic categories by predicting the class label of each pixel. We define five traversability categories Liang et al. (2025): road, sidewalk, vegetation, building, and others. Different types of robots have different traversable areas: wheeled robots are restricted to the sidewalk and road categories, whereas legged robots can also traverse stairs and curbs.

Given the segmented image, we overlay the set of candidate trajectories $\mathcal{T}$, generated by the CVAE-based trajectory generator, onto the image and evaluate their traversability scores. First, we use the Bresenham algorithm Bresenham (1998) to convert trajectory waypoints into connected pixels in the image. Then, the semantic traversability score s_τ is computed for each trajectory as the ratio of pixels falling in traversable areas to the total number of pixels along the trajectory.

In this work, we decouple the traversability analysis from social and traffic decision-making. It is reasonable because the physical traversability is the hard constraints that the robot will never moves into the non-traversable areas, and social and traffic navigation determines which trajectory is proper. So, unlike the original MOSU, we use these two complementary components to select trajectories only in traversable terrain. For the generated N trajectories, given the c_τ^i denotes the confidence score from geometric information, s_τ^i represents a semantic traversability score for each trajectory $i \in [1, N]$, we calculate the traversability of all the trajectories as

$$t_\tau^i = \beta_1 c_\tau^i + \beta_2 s_\tau^i; \qquad \mathcal{T}_s = \{\tau_i \in \mathcal{T} \mid t_\tau^i > k\}, \tag{4}$$

where $\beta_{1,2} > 0$ are weights of the components, and $\mathcal{T}_s$ represents the selected trajectories by traversability analysis.

VLM-retrievable Local Cost Map

We construct a local cost map that associates open-vocabulary semantic information with the geometry and motion of the surrounding environment, as illustrated in Fig. 4. The map represents traffic or social components in the environment, including crosswalks and traffic lights, together with the positions and velocities of nearby cars, pedestrians, and bicycles. These attributes provide evidence for social and traffic decisions beyond the physical traversability

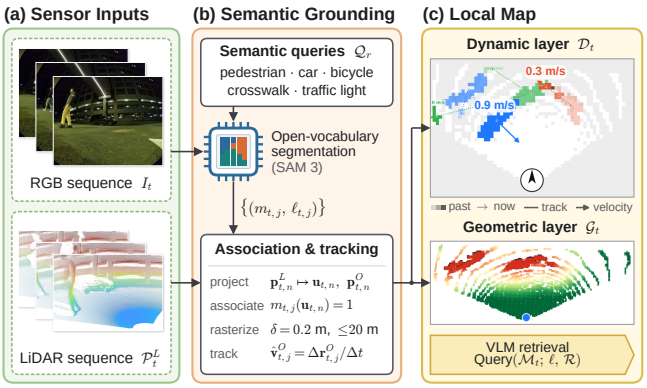


Figure 4. Construction of the VLM-retrievable local map. VLM provides robot-specific language queries for semantic segmentation of consecutive RGB images. Then the corresponding Lidar points are used to construct the 2D local map to provide the geometric information of the semantic objects, and calculate the motion information of the objects. The constructed geometric and dynamic layers allow VLMs to retrieve relevant information by semantic category and spatial location for social context understanding.

assessments described above. By exposing the traffic and social information through queryable function interfaces, the map allows VLMs to retrieve relevant environmental measurements and interpret them together with visual context.

At the beginning of each experiment, a language model generates a set of semantic queries, $\mathcal{Q}_r$, based on the robot's type and mobility capabilities. The query set remains fixed during each run. Given an RGB image I_t, the open-vocabulary segmentation model produces

$$\mathcal{S}_t = F_{\text{seg}}(I_t; \mathcal{Q}_r) = \{(m_{t,j}, \ell_{t,j})\}_{j=1}^{J_t}, \tag{5}$$

where $m_{t,j}$ is the binary mask of segment j, $\ell_{t,j}$ is its semantic label, and J_t is the number of detected segments. We use SAM3 Carion et al. (2026), consistent with the preceding traversability analysis. The queries include traffic-related concepts such as "traffic light," "car," "pedestrian," and "bicycle," together with additional concepts relevant to the robot platform.

To recover metric geometry, we associate each segmented image with the synchronized LiDAR point cloud $\mathcal{P}_t^L = \{\mathbf{p}_{t,n}^L\}$, expressed in the LiDAR frame L. We project the points onto the image plane to get the distances of the segments. Let $\mathbf{K}$ denote the camera intrinsic matrix, and let $(\mathbf{R}_{CL}, \mathbf{b}_{CL})$ denote the calibrated transformation from the LiDAR frame to the camera frame. A LiDAR point is projected into the rectified image as

$$\mathbf{u}_{t,n} = \pi\big(\mathbf{K}\left(\mathbf{R}_{CL}\mathbf{p}_{t,n}^L + \mathbf{b}_{CL}\right)\big), \tag{6}$$

where $\pi([x, y, z]^\top) = [x/z, y/z]^\top$. Points with valid positive depth and projections within the image are associated with the masks containing their projected pixels.

The associated points are then transformed into the odometry frame O:

$$\mathbf{p}_{t,n}^O = \mathbf{R}_{OL,t}\mathbf{p}_{t,n}^L + \mathbf{b}_{OL,t}, \tag{7}$$

where $(\mathbf{R}_{OL,t}, \mathbf{b}_{OL,t})$ is obtained from the robot's odometry pose and the calibrated LiDAR mounting transformation.

Expressing consecutive observations in the same odometry frame accounts for the robot's motion when comparing segment positions. For each segment, its associated point set and representative position are

$$\begin{aligned}\mathcal{P}_{t,j}^{O} &= \left\{\mathbf{p}_{t,n}^{O} \mid m_{t,j}(\mathbf{u}_{t,n}) = 1\right\}, \\ \mathbf{r}_{t,j}^{O} &= \frac{1}{|\mathcal{P}_{t,j}^{O}|} \sum_{\mathbf{p}\in\mathcal{P}_{t,j}^{O}} \mathbf{p}.\end{aligned} \tag{8}$$

Here, $\mathbf{r}_{t,j}^{O}$ is the centroid of the observed segment points. The full point set represents its observed spatial extent. Segments without valid LiDAR support do not receive a metric position estimate.

We maintain observations within $20\,\mathrm{m}$ radius around the robot, and discretize them into a local grid with resolution $\delta = 0.2\,\mathrm{m}$. Let $\mathbf{o}_t^O \in \mathbb{R}^2$ denote the grid origin in the odometry frame. The cells associated with segment j are

$$\Omega_{t,j} = \left\{ \left\lfloor \frac{\mathbf{p}_{xy}^{O} - \mathbf{o}_t^{O}}{\delta} \right\rfloor \,\middle|\, \mathbf{p}^{O} \in \mathcal{P}_{t,j}^{O} \right\}, \tag{9}$$

where $\mathbf{p}_{xy}^{O}$ contains the horizontal(xy) coordinates. This projection preserves the observed layout of an extended region, instead of representing it only by its centroid.

Motion information is estimated from the displacement of corresponding segments across consecutive observations. Let $a_t(j)$ identify the segment in the previous frame associated with segment j at time t. We associate each segment with the nearest segment of the same semantic class. Given an observation interval Δt, the object velocity estimate is

$$\hat{\mathbf{v}}_{t,j}^{O} = \frac{(\mathbf{r}_{t,j}^{O})_{xy} - (\mathbf{r}_{t-1,a_t(j)}^{O})_{xy}}{\Delta t}. \tag{10}$$

A velocity estimate is provided only when a valid temporal correspondence and sufficient geometric observations are available; otherwise, velocity is marked as unknown.

The semantic and motion attributes are organized into geometric and dynamic layers,

$$\begin{aligned}\mathcal{G}_t(\mathbf{a}) &= \left\{(j, \ell_{t,j}, \mathbf{r}_{t,j}^{O}) \mid \mathbf{a} \in \Omega_{t,j}\right\}, \\ \mathcal{D}_t(\mathbf{a}) &= \left\{(j, \hat{\mathbf{v}}_{t,j}^{O}) \mid \mathbf{a} \in \Omega_{t,j}\right\},\end{aligned} \tag{11}$$

where $\mathbf{a}$ denotes a grid cell. These layers associate semantic regions with their measured geometry and estimated motion.

Finally, we expose the map through function interfaces that support retrieval by semantic type, spatial location, or their combination. For a requested semantic label ℓ and a spatial region $\mathcal{R}$ expressed as a set of grid cells, a combined query is written as

$$\mathrm{Query}(\mathcal{M}_t; \ell, \mathcal{R}) = \left\{(\mathbf{r}_{t,j}^{O}, \hat{\mathbf{v}}_{t,j}^{O}) \mid \ell_{t,j} = \ell,\ \Omega_{t,j} \cap \mathcal{R} \neq \emptyset\right\}, \tag{12}$$

where $\mathcal{M}_t$ denotes the local map with its position, semantic, and dynamic information. We then convert them to the robot's frame. The returned measurements provide evidence for the subsequent two-stage reasoning process, which combines them with visual context and route guidance to select a trajectory from $\mathcal{T}_s$.

Two-stage Trajectory Selection

Given the traversable candidate set $\mathcal{T}_s$, we design a two-stage trajectory selection process to balance the computational cost of VLM-based reasoning with the need to recognize context-dependent navigation requirements. VLMs can interpret social and traffic cues using contextual knowledge, but detailed analysis of image sequences and repeated information retrieval can introduce substantial computation and latency. Meanwhile, manually specified detection rules may not cover the diverse interactions and unfamiliar conditions encountered during outdoor navigation. We therefore use a VLM in the first stage to examine the current observation and identify whether additional information is needed, allowing the observed context to guide further investigation. When the available information is sufficient, the system selects a trajectory from $\mathcal{T}_s$ according to the routed subgoal. Otherwise, the second stage analyzes consecutive RGB observations and retrieves relevant geometric and dynamic information from the local map to support trajectory selection and provide velocity mode. This architecture reserves detailed reasoning for situations identified as requiring further investigation, without requiring every triggering condition to be explicitly specified in advance.

In the first stage, candidate trajectories are projected to the image plane and labeled by numbers from right to left Liang et al. (2026). Then the projected trajectories are evaluated according to their progress toward the subgoal. Let $\mathbf{g}_t^B$ denote the routed subgoal expressed in the robot's current coordinate frame, and let $\mathbf{x}_{i,\mathrm{end}}^B$ denote the endpoint of candidate trajectory τ_i in the same frame. For a nonempty candidate set, the default trajectory is selected as

$$\tau_t = \arg\min_{\tau_i \in \mathcal{T}_s} \left\| \mathbf{x}_{i,\mathrm{end}}^{B} - \mathbf{g}_t^{B} \right\|_2^2. \tag{13}$$

This selection uses the geometric and semantic traversability assessments already incorporated into $\mathcal{T}_s$.

Before executing the selected trajectory, a VLM examines the current RGB observation together with the subgoal and candidate trajectories to determine whether additional scene information is needed:

$$(b_t, \mathcal{U}_t) = F_{\mathrm{screen}}\left(I_t, \mathbf{g}_t^{B}, \mathcal{T}_s\right), \tag{14}$$

where $b_t \in \{0, 1\}$ indicates whether to invoke the second stage, and $\mathcal{U}_t$ describes the information to be investigated. For example, the first stage may identify a pedestrian interaction, a traffic signal relevant to the heading route, or a temporary restriction that is not captured by physical traversability alone. The resulting request specifies the relevant scene elements and the unresolved question, such as whether the pedestrian in front is blocking the road or whether the bicycles or cars in traffic on the left are yielding. When no further investigation is requested, the system selects $\tau_t^{(1)}$.

When $b_t = 1$, the second stage receives the information requests and a sequence of recent H frames of RGB observations, $\mathcal{I}_t^H$. The image sequence provides temporal context for interpreting actions and changes in the scene. Based on $\mathcal{U}_t$ and these observations, the second stage selects semantic and spatial queries for the local map. Let $\mathcal{Q}_t^{\mathrm{map}}$

denote the requested combinations of semantic labels and spatial regions. The retrieved evidence is

$$\mathcal{Z}_t = \bigcup_{(\ell,\mathcal{R})\in\mathcal{Q}_t^{\text{map}}} \text{Query}\left(\mathcal{M}_t;\ell,\mathcal{R}\right), \tag{15}$$

For social interactions, the second stage interprets the consecutive images together with the retrieved positions and velocities of relevant pedestrians, cyclists, or vehicles. The images provide evidence about gestures, activities, and interaction context, while the map provides metric information about surrounding geometry and motion. Their combination supports distinctions that affect trajectory selection and velocity selection, such as whether a nearby road user is continuing toward the robot's path, moving away, or slowing near a crossing. These observations inform the interpretation of an interaction without treating motion alone as definitive evidence of intent. For traffic conditions, the second stage additionally interprets the state of the relevant traffic signal and its relationship to the intended crossing. Signal indications and any visible countdown information are considered together with the crossing geometry and the motion of nearby road users. The map supplies the spatial relationship of surrounding traffic, while the RGB observations provide the visual information needed to interpret the signal state. For irregular conditions, the second stage examines scene details that may change whether an otherwise traversable trajectory is appropriate. Examples include temporary barriers, construction activities, warning signs, and access restrictions. It uses the image sequence to interpret these cues. A scene may involve several of these conditions simultaneously, in which case their evidence is considered jointly.

The second stage combines the retrieved evidence, visual context, and subgoal information to select the final motion:

$$\tau_t, \mathbf{v}_t = F_{\text{select}}\left(\mathcal{T}_s, \mathbf{g}_t^B, \mathcal{I}_t^H, \mathcal{Z}_t, \mathcal{U}_t\right), \tag{16}$$

where $\mathbf{v}_t$ indicates the velocity mode, either stop, slow, or regular velocity. The selection considers progress toward the subgoal together with the social, traffic, and contextual implications of each candidate. The selected trajectory and velocity mode are passed to the motion controller, as shown in Fig. 2, and the selection process is repeated as new observations become available.

Experiments

We evaluate the navigation capabilities inherited from MOSU and the improvements introduced in MOSU2 through two groups of experiments. The first evaluates MOSU's traversability assessment and the ability of a single VLM agent to interpret social and traffic cues from an RGB image when ranking candidate trajectories. We retain selected results from the conference paper Liang et al. (2026) to establish the performance of this underlying navigation framework. The second compares MOSU2 with MOSU and controlled variants to examine how retrieving geometric and dynamic information from the local map affects navigation decisions and task completion. It also evaluates whether two-stage trajectory selection reduces reasoning overhead while recognizing and handling situations that require further investigation, including social interactions, traffic conditions, and irregular scenarios. Together, these experiments assess navigation performance, contextual understanding, and computational latency.

Experimental Setup

We conduct experiments on three robot platforms, as shown in Fig. 1: Husky, Unitree Go2, and Diablo robots, representing wheeled, legged, and wheeled-legged robots, respectively. Each platform is equipped with calibrated RGB cameras and 3D LiDAR for perceiving the surrounding environment, as well as GPS for global guidance.

Experiments are performed on a computer with an Intel i9 CPU (31 GB RAM) and an Nvidia RTX 3060 GPU (6.4 GB VRAM). We use ChatGPT-realtime Wu et al. (2023) or Gemini Team (2024) for visual-context interpretation and trajectory evaluation, with inference performed in the cloud through API access. The trajectory generator and motion planner operate at 50Hz and 100Hz, respectively. For MOSU2, the local map is updated at 2Hz with a 20 ⇳ radius around the robot, and the second reasoning stage receives $H = 5$ consecutive RGB images spanning 2.5 seconds.

Evaluation of MOSU

The experiment is designed to evaluate the efficacy of the system in two aspects: (1) the performance of each component, including point-cloud-based trajectory generation, semantic traversability analysis, and VLM's social understanding capability; (2) Overall MOSU system performance in real-world long-range navigation compared with other approaches.

We first summarize the trajectory selection mechanism used in MOSU Liang et al. (2026), which provides the baseline for evaluating the extensions introduced in MOSU2. MOSU combines geometric trajectory generation, semantic traversability assessment, and contextual evaluation by a single VLM agent. At each decision step, candidate trajectories are projected onto the current RGB image and assigned numerical labels according to their endpoint positions in the image. The agent receives this annotated image together with the relative position of the current subgoal and navigation instructions concerning pedestrian interactions, traffic rules, and robot-specific traversability. It returns a ranked list of trajectories and a velocity mode based on the observed scene.

The VLM ranking is converted into a score r_τ^i, with higher-ranked trajectories receiving larger scores. MOSU then selects a trajectory by combining this score with geometric confidence (c_τ^i), semantic traversability (s_τ^i), and progress toward the subgoal (g_τ^i):

$$\tau^* = \arg\max_{\tau_i\in\mathcal{T}} \left(\alpha_c c_\tau^i + \alpha_s s_\tau^i + \alpha_r r_\tau^i + \alpha_g g_\tau^i\right), \tag{17}$$

where $\mathcal{T}$ denotes the candidate trajectory set, c_τ^i is the geometric confidence score, s_τ^i is the semantic traversability score, and g_τ^i favors trajectories approaching the current subgoal. The coefficients α_c, α_s, α_r, and α_g weight the respective components. The selected trajectory and velocity mode are passed to the motion controller. This mechanism uses a single RGB image for VLM-based contextual

evaluation, without the map retrieval or conditional second-stage reasoning introduced in MOSU2.

We first qualitatively evaluate each component, including traversability analysis using geometric trajectory generation, semantic information, and VLM-based social and traversability understanding of the environment, and then we quantitatively evaluate whether this combination produces traversable trajectories that approach the destination. The quantitative evaluation uses the GND dataset Liang et al. (2025), which contains observations from ten campuses spanning urban and rural environments. The evaluated scenarios include narrow passages, off-road terrain, crosswalks, and pedestrian interactions, as illustrated in Fig. 5. On this dataset we compare MOSU against MTG Liang et al. (2024a), DTG Liang et al. (2024b), and VL-TGS Song et al. (2025b), NoMaD Sridhar et al. (2024), ViNT Shah et al. (2023), and PIVOT Nasiriany et al. (2024). The results retained here use the original MOSU configuration, including Gemini as the VLM backend.

The metrics in Table 1 are calculated as following:

- **Traversability:** We overlay the generated trajectory onto the traversability map from the GND dataset Liang et al. (2025). The traversability score is then calculated as the percentage of fully traversable waypoints over the entire trajectory length.
- **Distance to Target:** $1 - \frac{d_\tau - d_o}{|\tau|}$, where d_τ and d_o represent the distances between the target and the current trajectory and between the target and the optimal ground-truth trajectory, respectively. $|\tau|$ denotes the length of the generated trajectory.

Quantitative results. Tab. 1 reports the retained comparison. MOSU achieves a traversability score of 77%, exceeding the strongest competing result, PIVOT's 70%, by seven percentage points, corresponding to a relative improvement of 10%. MOSU also achieves the highest distance-to-target score of 73%, compared with 70% for VL-TGS and 69% for PIVOT. These results indicate that the combined evaluation improves trajectory traversability while maintaining progress toward the destination under the reported benchmark. The reported inference time for MOSU is $2.30, \mathrm{s}$, matching PIVOT and closely matching VL-TGS at $2.31, \mathrm{s}$ with Gemini. The remaining methods report inference times between $0.01, \mathrm{s}$ and $0.69, \mathrm{s}$. The results show that the VLM is the bottleneck in inference time. Therefore, in MOSU2 we leverage the ChatGPT-realtime with 1Hz inference time for each analysis.

Qualitative results. As shown in Fig. 7 (a), geometry-based trajectory generation Liang et al. (2024a) performs well in scenarios where geometric structures are easily detected. However, when the elevations of off-road areas and sidewalks are similar, it often fails, as shown in (c). In Fig. 7 (a) and (b), lighter colors indicate higher scores in both geometric confidence and color-based traversability. As shown in (d), semantic segmentation also encounters out-of-distribution scenarios, where large ground areas cannot be properly segmented. Therefore, geometric and semantic information must be combined for reliable traversability analysis, and MOSU2 inherits this design to filter candidate trajectories. As shown in Table 1, our approach achieves the best traversability among all methods, and the generated trajectories lead to the target closely.

As shown in Fig. 6, beyond traversability analysis, VLMs process labeled trajectories and analyze social cues from the images. In the first row, the VLM ranks the trajectories as [3, 0, 1, 2, 4], given the target at the right front. It suggests following trajectory 3 at normal speed, while other trajectories should be taken at a slower speed when encountering humans. The ranking for the second row is [4, 5, 0, 1, 2, 3] with normal speed. In the experiment, we observed that VLMs take a significant amount of time to process, as shown in Table 1, but they demonstrate strong capability in understanding social cues.

These examples provide qualitative evidence of contextual trajectory ranking. They do not quantify social-cue recognition accuracy or establish the reliability of decisions that depend on accurately measured object motion. The following experiments therefore examine whether access to local geometric and dynamic information improves navigation under social, traffic, and irregular conditions, and whether the two-stage architecture provides these capabilities with lower reasoning overhead.

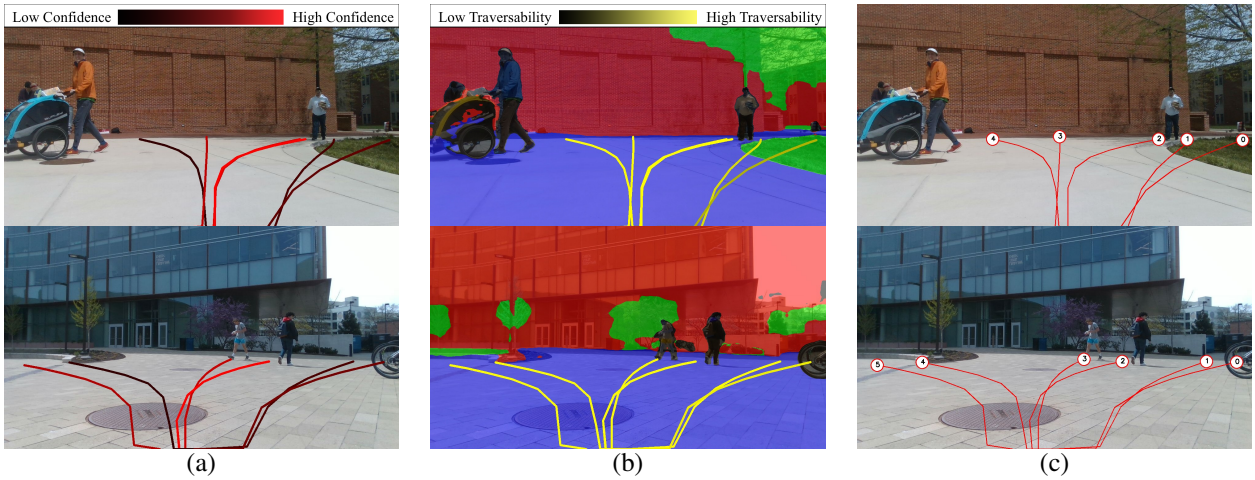
Evaluation of MOSU2

We compare MOSU2 with MOSU to evaluate navigation decisions under social, traffic, and irregular environmental conditions. As shown in Fig.8, the evaluation covers seven scenarios: stop gestures; construction sites; signalized crosswalks with cars, pedestrians and bicycles; roads shared with vehicles, cyclists, or pedestrians; and unsignalized roads. These scenarios examine complementary capabilities, including interpreting human gesture, following temporary restrictions, and coordinating robot motion with traffic signals and surrounding road users. MOSU performs contextual trajectory evaluation using a single RGB image and one VLM agent, whereas MOSU2 use two-stage strategy and the 2nd stage extends this mechanism through conditional analysis of recent RGB observations and retrieval of local geometric and dynamic information. We report navigation success rate (Nav. Success Rate), reasoning latency in seconds (Latency), and condition-detection accuracy (Det. Accuracy). Table 2 summarizes all evaluated trials within each scenario.

Navigation performance. MOSU2 improves navigation success in four of the five evaluated scenario groups and matches MOSU in the baseline scenario. The unweighted mean navigation success rate increases from 41.2% for MOSU to 93.4% for MOSU2, representing an absolute improvement of 52.2 percentage points. At signalized crosswalks and on roads shared with vehicles, cyclists, or pedestrians, MOSU2 achieves 100% navigation success compared to 0% and 56% for MOSU, respectively. These dynamic settings require reasoning over traffic signal states and surrounding agent trajectories, where MOSU's single-frame visual scoring frequently leads to overly conservative stops or premature entry. In construction site environments featuring temporary signage and barriers, MOSU2 increases navigation success from 50% to 100%, demonstrating robust path selection around physical restrictions. For human stop gestures, navigation success increases from 0% to 67%,

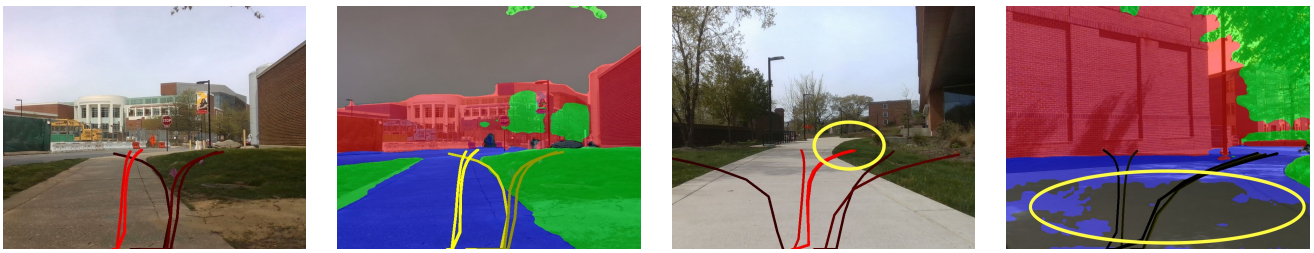
(a) Narrow Space (b) Off-road Terrain (c) Crosswalks (d) Social Scenarios

Figure 5. Complex Outdoor Scenarios: We evaluate the approach in large-scale, complex outdoor environments with various challenging scenarios, such as (a) narrow spaces, (b) areas with dense off-road vegetation, (c) traffic components (e.g., crosswalks), and (d) social situations involving pedestrians.



(a) (b) (c)

Figure 6. Qualitative Evaluation: Each column illustrates one of the three components in our trajectory generation pipeline, (a) CVAE-based multiple trajectory generation, (b) semantic segmentation for traversability analysis, and (c) VLM-based trajectory ranking in the outdoor scenarios. In (a) and (b), lighter colors indicate higher scores, confidence scores in (a) and traversability scores in (b). In (c), the VLM ranks trajectories based on social cues. The top example shows a ranking of [3, 0, 1, 2, 4]. The bottom example shows a ranking of [4, 5, 0, 1, 2, 3].



(a) (b) (c) (d)

Figure 7. Geometric and Semantic Analysis: Both geometric trajectory generation and semantic segmentation perform well in most cases, but there are also failure cases, as marked in yellow circles in (c) and (d). When the elevations of the lawn and sidewalk are similar, the geometric model struggles to perform accurately. Additionally, segmentation fails in out-of-distribution scenarios.

Method	Modality	Traversability (%) ↑	Distance to Target (%) ↑	Inference Time (s) ↓
PIVOT Nasiriany et al. (2024)	RGB + Language	70	69	2.30
ViNT Shah et al. (2023)	RGB	57	62	0.69
NoMaD Sridhar et al. (2024)	RGB	59	61	0.24
MTG Liang et al. (2024a)	Points	61	64	0.01
DTG Liang et al. (2024b)	Points	67	66	0.13
VL-TGS Song et al. (2025b)	Points, RGB, Language	65	70	2.31
MOSU (ours) Liang et al. (2026)	Points, RGB, Language	**77**	**73**	2.30

Table 1. Quantitative Evaluation: Our approach achieves the best Traversability and comparable Distance to Target.

reflecting improved responsiveness to social cues, though minor edge cases remain when gesture visibility is obscured. Both frameworks achieve 100% navigation success in the unsignalized-road baseline scenario, confirming that MOSU2 preserves performance on routine paths while substantially expanding capabilities in complex, dynamic settings.

Condition detection and navigation decisions. The unweighted mean condition-detection accuracy rises from 58.2% for MOSU to 91.4% for MOSU2, an increase of 33.2 percentage points. The most pronounced improvement occurs in the stop gesture scenario, where detection accuracy improves from 0% to 89%. Detection accuracy also increases from 41% to 93% at signalized crosswalks and from 50% to 89% in shared-road environments with moving vehicles, bicycles, and pedestrians. These gains confirm that combining temporal image sequences with queryable metric geometry and velocity estimates enhances the recognition of dynamic contextual conditions.

The construction site results highlight the crucial distinction between contextual condition detection and executable navigation decisions: both MOSU and MOSU2 achieve 100% detection accuracy, yet MOSU2 doubles navigation success from 50% to 100%. This disparity demonstrates that correct scene classification is insufficient on its own; effective autonomy requires grounding detected conditions in measured object motion to evaluate trajectory feasibility. Conversely, in the unsignalized-road baseline, MOSU2's detection accuracy drops slightly from 100% to 86% due to occasional false-positive escalation triggers in benign surroundings, yet navigation success remains at 100%. Thus, MOSU2 significantly improves overall context recognition and decision reliability without requiring perfect condition classification in every single frame.

Reasoning latency. The performance gains achieved by MOSU2 incur higher computational and reasoning latency. The unweighted mean decision latency increases from 1.74 s for MOSU to 4.82 s for MOSU2, representing approximately a $2.77\times$ increase over the baseline. MOSU2's latency ranges from 3.52 s in the stop gesture scenario to 5.75 s in the unsignalized-road scenario. This additional overhead originates from the multi-frame temporal RGB sequence analysis ($H = 5$ frames over 2.5 s) and the function-based retrieval of geometric and dynamic layers from the local map. While conditional escalation restricts second-stage deliberation to ambiguous or high-risk scenes, the results characterize an explicit trade-off between higher navigation reliability and increased decision latency during complex interactions.

Conclusion

We presented MOSU2, an extension of MOSU for long-range outdoor navigation that combines route guidance, traversability assessment, and contextual reasoning. Building on MOSU's GPS-based routing and multi-modal trajectory generation and evaluation, MOSU2 introduces a VLM-retrievable local map that associates semantic regions with measured geometry and estimated motion. This information complements consecutive RGB observations when the decision agent interprets social interactions, traffic conditions, and irregular situations. A two-stage multi-agent architecture determines when additional investigation is needed: in the first stage, a context agent screens the current scene and supports routine subgoal-directed navigation, while in the second stage, a decision agent retrieves relevant evidence and uses it to select a trajectory and velocity mode. Together, these components connect contextual understanding with environmental measurements and robot-specific traversability, while reserving detailed reasoning for situations identified as requiring further investigation.

In real-world experiments across five scenarios, MOSU2 achieved 100% navigation succesws in four scenarios and 67% in the stop-gesture scenario, whereas MOSU succeeded in 0–50% of the trials in the four scenarios that required contextual reasoning. MOSU2 detected the relevant conditions with 86–100% accuracy, at the cost of higher reasoning latency (3.5–5.8 s, compared with 1.6–1.9 s for MOSU).

Limitations and Future Work

MOSU2 depends on the quality of its geometric, semantic, and dynamic observations. Occlusions, sparse LiDAR returns, segmentation errors, and calibration or odometry inaccuracies can affect the local map. In particular, associating segments by semantic category and spatial proximity can produce incorrect correspondences when several nearby objects share the same category. Changes in the visible portion of an object can also shift its observed centroid and introduce errors into the velocity estimate. Future work will investigate more robust temporal association and motion estimation, together with uncertainty estimates that allow the reasoning process to distinguish reliable measurements from incomplete or ambiguous evidence.

The two-stage architecture also depends on the first stage recognizing situations that require further investigation. A missed cue can prevent the second stage from being invoked, while unnecessary escalation can increase computation and delay. Moreover, the first stage itself requires VLM inference, so conditional escalation does not eliminate reasoning overhead. Future work will explore uncertainty-aware escalation, lightweight screening models, and mechanisms for revisiting decisions as new observations arrive. Evaluation across a broader range of rare and unfamiliar situations will be necessary to characterize the trade-off between reasoning cost and missed contextual requirements.

Even when additional evidence is available, VLM interpretations can remain ambiguous or incorrect. Gestures, signal relevance, and vehicle behavior may admit several explanations, and motion measurements alone cannot establish another road user's intent. With reasoning latencies upto 5.75 s (Table 2), the scene can also change before a decision is executed. Future work will investigate explicit handling of conflicting evidence, checks on observation freshness, and independent verification of selected motions against updated geometric and dynamic observations.

Finally, trajectory selection is limi by the quality and coverage of the generated candidates. Small height

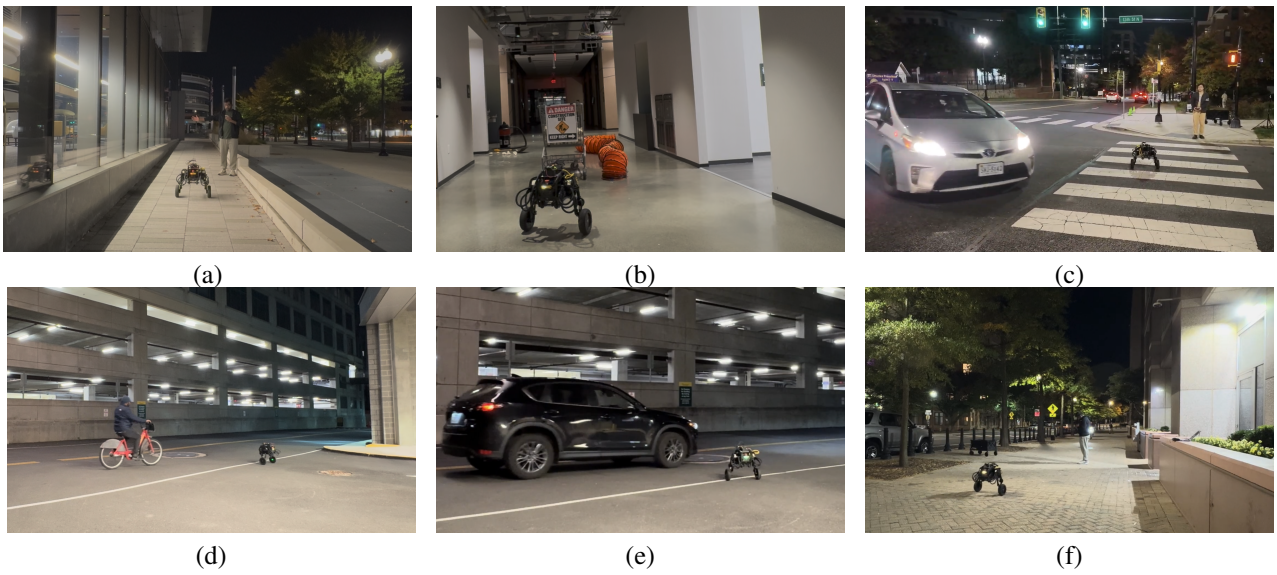


Figure 8. Representative social, traffic, and irregular navigation scenarios used to evaluate MOSU2: (a) responding to a pedestrian's stop gesture, (b) navigating a construction area with warning and directional signs, (c) crossing at a signalized intersection, (d) sharing the road with moving agents, illustrated by a cyclist, (e) intersecting with a vehicle, requiring the robot to analyze the car's velocity to determine if it is yielding, and (f) navigating an unsignalized road environment, serving as a baseline for comparisons. These scenarios require interpreting contextual cues and, where relevant, considering the geometry and motion of surrounding road users to determine appropriate robot behavior.

Table 2. Performance Comparison: MOSU vs. MOSU2 across diverse dynamic scenarios.

Scenario	MOSU			MOSU2		
	Nav. Success Rate (%)↑	Latency (s)↓	Det. Accuracy (%)↑	Nav. Success Rate (%)↑	Latency (s)↓	Det. Accuracy (%)↑
Stop gesture	0	**1.61**	0	**67**	3.52	**89**
Construction site	50	**1.68**	**100**	**100**	5.16	**100**
Signalized crosswalk	0	**1.90**	41	**100**	4.72	**93**
Road + Vehicles/Bikes/Peds	56	**1.785**	50	**100**	4.95	**89**
Unsignalized road	**100**	**1.74**	**100**	**100**	5.75	86

discontinuities, partially observed terrain, and unfamiliar environmental features can lead to inaccurate traversability assessments, and contextual reasoning cannot select an appropriate motion if it is absent from the candidate set. The map's semantic queries are also fixed during each run, limiting the categories available for retrieval. Future work will examine adaptive trajectory generation, online refinement of semantic queries, and recovery strategies when no suitable candidate is available, together with broader evaluation across robot platforms and outdoor conditions.

Acknowledgements

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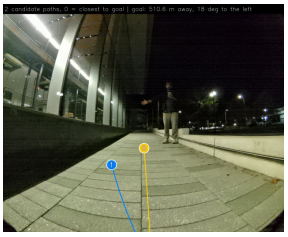
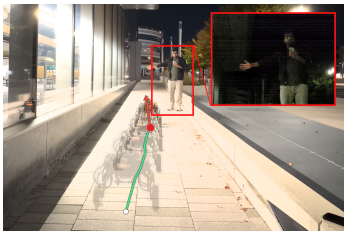
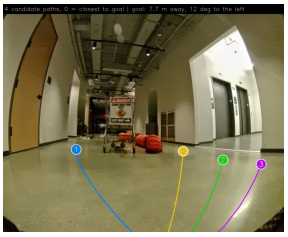
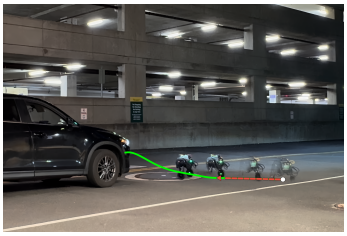
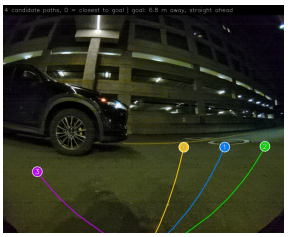
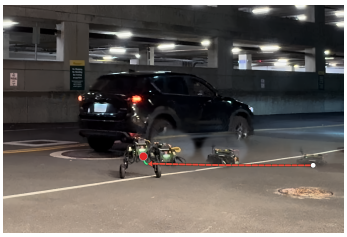
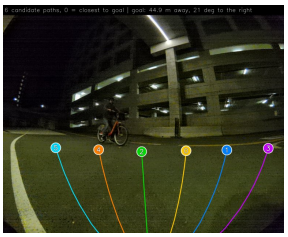
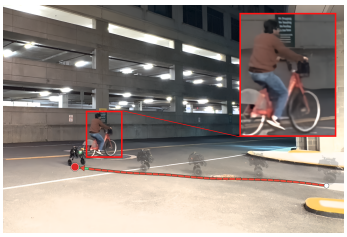
First-Person View	VLM Reasoning & Decision Output	Third-Person View (Execution)
	**Stage 1 Screening:** Pedestrian raising hand in path. Decision required on whether to stop. Escalating to Stage 2. **Stage 2 Decision:** Queried local map for pedestrian state. Detected active stop gesture and zero relative velocity ($v = 0.0$ m/s). Selected **Trajectory 3** with **Stop** velocity mode.	
	**Stage 1 Screening:** Pedestrian raising hand in path. Decision required on whether to stop. Escalating to Stage 2. **Stage 2 Decision:** Queried local map for pedestrian state. Detected active stop gesture and zero relative velocity ($v = 0.0$ m/s). Selected **Trajectory 3** with **Stop** velocity mode.	
figures/quality_social_traffic/1st/03.fpv.jpg	**Stage 1 Screening:** Pedestrian raising hand in path. Decision required on whether to stop. Escalating to Stage 2. **Stage 2 Decision:** Queried local map for pedestrian state. Detected active stop gesture and zero relative velocity ($v = 0.0$ m/s). Selected **Trajectory 3** with **Stop** velocity mode.	
	**Stage 1 Screening:** Pedestrian raising hand in path. Decision required on whether to stop. Escalating to Stage 2. **Stage 2 Decision:** Queried local map for pedestrian state. Detected active stop gesture and zero relative velocity ($v = 0.0$ m/s). Selected **Trajectory 3** with **Stop** velocity mode.	
	**Stage 1 Screening:** Pedestrian raising hand in path. Decision required on whether to stop. Escalating to Stage 2. **Stage 2 Decision:** Queried local map for pedestrian state. Detected active stop gesture and zero relative velocity ($v = 0.0$ m/s). Selected **Trajectory 3** with **Stop** velocity mode.	
figures/quality_social_traffic/1st/06.fpv.jpg	**Stage 1 Screening:** Pedestrian raising hand in path. Decision required on whether to stop. Escalating to Stage 2. **Stage 2 Decision:** Queried local map for pedestrian state. Detected active stop gesture and zero relative velocity ($v = 0.0$ m/s). Selected **Trajectory 3** with **Stop** velocity mode.	

Figure 9. Qualitative evaluation of MOSU2 across representative interactions: Left column displays the onboard egocentric camera observation (First-Person View). Middle column shows the textual reasoning output from the two-stage multi-agent engine, including screening decisions, map retrieval, and chosen motion. Right column shows the external observer view (Third-Person View) demonstrating physical execution.

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